Sachin kumar

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Hybrid Neural-Sparse Retrieval-Augmented Generation for Trustworthy Pharmacovigilance Conversational Intelligence

Sachin Kumar[1], [Supervisor Name][2]

[1] Department of Data Science, Delhi Technological University, Delhi – 110042, India.

[2] Department of [Department], Delhi Technological University, Delhi – 110042, India.

Email: [1] [email@dtu.ac.in] [2] [supervisor@dtu.ac.in]

ORCID Id: [1] [Author 1 ORCID] [2] [Author 2 ORCID]

Corresponding Author*: Sachin Kumar.

ABSTRACT:

Pharmacovigilance professionals must reconcile evidence from FAERS case reports, ICH/EMA regulatory guidelines, and biomedical literature residing in separate retrieval systems — a cross-source process that practitioners consistently identify as among the most time-consuming components of complex adverse event signal assessment [29]. Retrieval-Augmented Generation (RAG) reduces this burden by anchoring language model responses to explicitly retrieved passages, thereby limiting hallucination and enabling source attribution. To the best of our knowledge, no prior published study has systematically compared sparse, dense, and hybrid retrieval architectures on a pharmacovigilance corpus, or quantified how evaluation threshold selection distorts retrieval and generation metrics. We address both gaps by constructing a purpose-built corpus of 19,930 document chunks and evaluating three RAG architectures against 100 held-out pharmacovigilance queries. SimpleRAG applies BGE-large-en-v1.5 dense retrieval over a FAISS Flat IP index and forwards five passages to the generator. OptimisedRAG fuses BM25 with dense retrieval via Reciprocal Rank Fusion (RRF), applies MiniLM cross-encoder reranking and Maximal Marginal Relevance (MMR) diversification, and forwards ten passages. TurboRAG replaces the flat index with FAISS IVFFlat while retaining the full hybrid pipeline. At threshold $\tau = 0.65$, TurboRAG achieves Hit@5 = 1.000, NDCG@5 = 0.898, and ROUGE-1 = 0.471, leading across all 5 retrieval quality and all 7 generation quality metrics. SimpleRAG retrieves in 32 ms at Hit@5 = 0.960 with the highest Context Precision. API generation consumes approximately 1,090 ms regardless of retrieval architecture, reframing the cost-benefit case for pipeline complexity. Threshold sensitivity analysis shows retrieval metrics degrade up to 41% relative between $\tau = 0.65$ and $\tau = 0.75$ while generation quality holds stable (ΔROUGE-1 < 0.01) — an asymmetry with direct implications for PV evaluation standardisation. All reported metrics are point estimates on 100 queries from a single experimental run.

KEYWORDS: pharmacovigilance; retrieval-augmented generation; hybrid information retrieval; BM25; FAISS IVFFlat; cross-encoder reranking; maximal marginal relevance; NDCG; ROUGE; threshold sensitivity; adverse drug reaction

1. Introduction

A pharmacovigilance scientist assessing a hepatotoxicity signal for a novel kinase inhibitor must consult the ICH E2A causality framework, FAERS historical precedents, MedDRA Preferred Term mappings, and the current EU SmPC — each in a separate retrieval system. Evidence reconciliation across these sources is among the most time-intensive tasks in pharmacovigilance case assessment [29], [6], and the burden compounds as submission volumes grow.

Retrieval-Augmented Generation (RAG) [2] consolidates dispersed sources by prepending retrieved passages to a language model's input window, grounding responses in verifiable text and substantially limiting hallucination [3]. Uptake in clinical NLP is accelerating: Li et al. evaluated GPT-3.5, GPT-4, and Claude 2 for automated PV literature screening and reported 97% sensitivity but only 67% specificity [28], a precision gap that motivates hybrid sparse-dense retrieval with more selective evidence grounding. Ball et al. demonstrated that algorithmic trustworthiness — not performance metrics alone — governs clinician acceptance of AI tools at the FDA [29], motivating source-attributable RAG over parametric generation. Yang et al. showed that a knowledge-graph-augmented

RAG framework raised ROUGE-L by 18% over baseline generation in biomedical QA [4], confirming that retrieval quality governs generation fidelity.

Pharmacovigilance retrieval blends exact regulatory identifiers — MedDRA Preferred Terms, ICH section numbers, FAERS case IDs — with clinical paraphrases, challenging both pure lexical and pure dense retrieval. Prior work has evaluated dense RAG for PV literature review [5] and AI-based signal management [6], but to the best of our knowledge no controlled comparison of sparse, dense, and hybrid retrieval architectures on a PV corpus has been published, nor has evaluation threshold choice been studied in this context.

Three research questions guide this work:

RQ1. Does combining sparse (BM25) and dense (BGE) retrieval with cross-encoder reranking improve pharmacovigilance RAG performance over a dense-only baseline?

RQ2. How does the embedding similarity threshold τ affect retrieval versus generation evaluation metrics, and are these effects decoupled?

RQ3. Does retrieval or generation dominate end-to-end response latency, and what are the deployment implications?

This paper delivers four contributions:

1. Three RAG architectures — dense-only SimpleRAG, hybrid OptimisedRAG, and IVFFlat-index TurboRAG — evaluated on an identical PV corpus with a shared generator (context-budget difference between systems is explicitly reported as a confound).
2. First threshold sensitivity study for PV-RAG, to our knowledge: tightening τ from 0.65 to 0.75 collapses retrieval metrics by up to 41% relative while generation quality holds stable.
3. Latency decomposition establishing API generation, not retrieval, as the dominant time cost — reframing the case for hybrid pipeline investment.
4. A 500-QA pharmacovigilance benchmark spanning four query categories and three document types, to be released on acceptance.

2. Related Work

NLP in pharmacovigilance has progressed from rule-based NER to neural transformers. Luo et al. surveyed the field through 2017, noting that semantic and syntactic modelling increasingly governs ADE detection accuracy [7]. Murphy et al. reviewed 29 ADE detection studies and found that aggregate F1 scores overstate performance on the ADE entity class due to class imbalance [8]. McMaster et al. showed domain-adaptive pretraining of DeBERTa on 1.1 million clinical documents raised ADR detection ROC-AUC to 0.955 [9]; Wu et al. applied the same principle to FDA drug labelling via RxBERT, reaching 86.5 F1 on NIST TAC [10]. Silverman et al. adapted UCSF-BERT for serious adverse event identification from outpatient clinical notes in IBD cohorts, achieving 88–92% accuracy on drug-SAE pair detection [30]. Li et al. confirmed BERT-family models as the leading approach for clinical ADE extraction [11]; Pilipiec et al. documented methodological variability across social-media PV NLP [12]; Warner et al. found ML-based signal management outperforms traditional disproportionality statistics but lacks transparency [6]; Painter et al. showed retrieval quality is the binding constraint on PV-RAG fidelity, examining dense-only retrieval without threshold sensitivity [5].

In information retrieval, BM25 [13] achieves robust sparse matching on out-of-distribution text [14]. Dual-encoder dense retrieval [15], [16] enables semantic matching; BGE-large-en-v1.5 [17] leads the BEIR zero-shot benchmark [14]. FAISS IVFFlat [18] provides approximate nearest-neighbour search with minimal recall degradation. RRF [19] fuses ranked lists without score normalisation; cross-encoder reranking [20] achieves higher accuracy than bi-encoder similarity; MMR [21] produces diverse result sets. Lewis et al. formalised RAG [2]; Gao et al. classified RAG into naive, advanced, and modular paradigms [22]; Ji et al. established retrieval grounding as a primary hallucination mitigation [3]. The RAGAS framework [23] introduced context recall and precision as passage-level metrics; the present work adopts these but uses embedding-based relevance labels rather than

RAGAS's original LLM-based assessment — a deviation that introduces the circularity limitation discussed in Section 4.

3. System Design

3.1 Corpus Construction

Three document categories were ingested: anonymised FAERS case narratives, PV reference textbooks, and ICH/EMA regulatory guidelines. A 500-token sliding window with 50-token overlap produced 19,930 passages (mean 387 tokens). The window size encompasses complete ICH E2A subsections (300–450 tokens average) without splitting drug-reaction co-occurrences; the 50-token overlap preserves sentences that span chunk boundaries. Section headings were prepended to guideline chunks, embedding regulatory context such as 'ICH E2A Section 3.2 Causality Assessment:' in the passage encoding for improved regulatory query retrieval. Chunk size was not optimised through ablation and remains an untuned hyperparameter; sensitivity to 256- and 750-token windows is a priority for future work.

All 19,930 passages were encoded offline by BGE-large-en-v1.5 [17] on a Tesla T4 in float16 (equation 1). BGE-large-en-v1.5 was chosen because it achieves the highest NDCG@10 on the BEIR zero-shot benchmark [14], offers 1024-dimensional discrimination between clinically adjacent terms, and matches or exceeds domain fine-tuned models in zero-shot PV retrieval [9], [10]. Queries receive an instruction prefix (equation 2) [17] that biases the encoding toward asymmetric retrieval. L2-normalisation makes FAISS inner products equivalent to cosine similarities.

$$\hat{d}_i = BGE(t_i) / \|BGE(t_i)\|_2 \in \mathbb{R}^{1024} \quad (1)$$

$$\hat{q} = BGE(prefix + q) / \|BGE(prefix + q)\|_2,\ prefix = \text{'Represent this sentence for searching relevant passages:'} \quad (2)$$

3.2 System 1: SimpleRAG

SimpleRAG returns the top-5 passages with the highest cosine similarity to $\hat{q}$ from a FAISS Flat IP scan of all 19,930 embeddings. Exact search at O(N) complexity completes in ~4 ms on the T4 [18]. No reranking or diversification is applied. This is the minimal-latency, minimal-complexity baseline, and the five passages are forwarded directly to the generator.

3.3 System 2: OptimisedRAG

A five-stage hybrid pipeline. (1) BM25 [13] (equation 3) with $k_1 = 1.5$, $b = 0.75$ — the standard Okapi defaults [13], not tuned — retrieves 100 candidates B. (2) FAISS Flat IP returns 100 candidates E. (3) RRF [19] (equation 4) merges both lists. The 0.4/0.6 BM25-to-dense weight was chosen based on the general principle that higher dense weighting benefits semantic QA tasks [14], [19]; systematic grid-search over fusion weights on a validation partition is recommended for production deployments. (4) ms-marco-MiniLM-L-6-v2 [20] reranks the top-100 RRF-merged candidates (equation 5), executed with batch_size = 1 on the T4 (~11.4 ms per pair). (5) MMR [21] (equation 6) with $\lambda = 0.75$ — an untuned default — selects ten diverse passages. The ten passages are forwarded to the generator.

$$BM25(q,d) = \Sigma[t \in q]\ \log((N - n\_t + 0.5)/(n\_t + 0.5) + 1) \times (f(t,d)(k_1+1))/(f(t,d) + k_1(1 - b + b \cdot |d|/avgdl)) \quad (3)$$

$$s\_RRF(d) = 0.4/(60 + r\_B(d)) + 0.6/(60 + r\_E(d)) \quad (4)$$

$$\hat{s}(q,d) = CrossEncoder([q \,||\, d]) \quad (5)$$

$$d^* = argmax\_\{d\_i \notin S\}\ [\lambda \cdot sim(\hat{q}, \hat{d}_i) - (1-\lambda) \cdot max\_\{d\_j \in S\}\ sim(\hat{d}_i, \hat{d}_j)],\ \lambda = 0.75,\ k = 10 \quad (6)$$

3.4 System 3: TurboRAG

TurboRAG replaces the flat dense index with FAISS IVFFlat [18]: k-means partitions 19,930 embeddings into n_list = 128 Voronoi cells; at query time n_probe = 16 cells are exhaustively searched (equation 7). n_list = 128 was chosen as the nearest power of 2 below $\sqrt{N} \approx 141$, following FAISS convention [18]. With n_probe = 16, 12.5% of the index is searched at an 8× reduction in search space; recall@100 deviates from exact search by under 0.5%. All downstream stages — BM25 retrieval, RRF, cross-encoder reranking of the top-100 RRF-merged candidates, and MMR — are identical to OptimisedRAG.

$$E\_IVF = top\text{-}100(\{\hat{d}_i : \hat{d}_i \in C\_j,\ j \in top\text{-}16\ cells\}),\quad n\_list=128,\ n\_probe=16 \qquad (7)$$

TurboRAG's consistent quality advantage over OptimisedRAG (e.g., NDCG@5 = 0.898 vs. 0.805 at $\tau = 0.65$) is counterintuitive given that both systems apply identical RRF, cross-encoder, and MMR stages to approximately 100 RRF-merged candidates. We hypothesise that Voronoi partitioning implicitly concentrates semantically similar passages within clusters, so that searching the 16 nearest cells may yield a more locally coherent candidate pool than a global Flat IP scan, providing the cross-encoder with a more focused reranking input. This interpretation remains unconfirmed; an ablation experiment substituting IVFFlat for Flat IP with all other variables held constant is needed to verify or refute it, and is a priority for future work.

Regarding latency: both OptimisedRAG and TurboRAG collect ~100 BM25 candidates and ~100 dense candidates before RRF, and both rerank approximately 100 RRF-merged candidates through the cross-encoder (batch_size = 1). TurboRAG's higher retrieval latency (2,394 ms vs. 1,195 ms) is therefore not attributable to doubled cross-encoder calls. Instead, the additional overhead likely arises from IVFFlat cell traversal and non-contiguous memory access across 16 Voronoi partitions, compared to the sequential contiguous scan of Flat IP. Detailed per-component profiling within the retrieval stage would be needed to precisely characterise this overhead.

3.5 Generation and Prompting

All three systems share Amazon Bedrock Nova Lite (model ID: *amazon.nova-lite-v1:0*, T = 0.05, max_tokens = 800). At T = 0.05, decoding is near-deterministic: temperature is sufficiently low that stochastic sampling introduces negligible variance for clinical text generation, though strictly T = 0 is required for full determinism [3]. The 800-token output accommodates ICH E2A assessments (300–600 words typical). Context budgets differ by design: SimpleRAG forwards 5 passages; OptimisedRAG and TurboRAG forward 10 after MMR. This difference is explicit: generation metrics between SimpleRAG and the hybrid systems reflect both retrieval quality and context-size effects jointly and cannot be attributed to retrieval architecture alone. Controlling for context budget is the single most important experiment for future work. The grounding prompt for all systems: *'Answer using only information from the provided context passages. State if context is insufficient. Do not hallucinate.'* [2]

[Figure 1 — Insert image here]

Figure 1. End-to-end pipeline. Five stages: (1) corpus ingestion and 500-token chunking, (2) BGE-large-en-v1.5 offline embedding and index construction, (3) three parallel retrieval columns — SimpleRAG (blue), OptimisedRAG (green), TurboRAG (amber), (4) shared Amazon Bedrock Nova Lite generation, and (5) 17-metric evaluation harness.

4. Evaluation Framework

A passage d_i is judged relevant when $sim(\hat{d}\_i, \hat{d}^*) \geq \tau$ for any ground-truth passage d*. Evaluating at $\tau \in \{0.65, 0.70, 0.75\}$ brackets the empirical range where BGE similarities cluster for paraphrastic FAERS descriptions of the same clinical event, avoiding single-threshold artefacts. All relevance labels and context metrics are derived from the same BGE encoder used for retrieval — a circularity that may bias results toward dense-retrieval systems. Context Recall and Precision metrics are inspired by RAGAS [23] but use embedding-based labels rather than RAGAS's original LLM-based assessment, constituting an implementation deviation that trades faithfulness for computational tractability. Human-judged relevance labels would mitigate both limitations. Five retrieval metrics are computed at $K \in \{1, 3, 5, 10\}$: Hit@K, Recall@K, NDCG@K [24] (equation 8), MRR, and MAP; Table 1 reports K = 5 as the primary cutoff because five passages is the common-practice retrieval budget in RAG evaluation at this corpus scale. Seven generation metrics are reported: ROUGE-1/2/L [25], BLEU-4 [26], METEOR [27], Token F1, and BGE cosine semantic similarity. Latency is decomposed into retrieval and generation components, adding 3 efficiency metrics for a total of 17. **All metrics are point estimates from a single experimental run on 100 test queries; confidence intervals and significance tests are outside the scope of this study and are a recommended extension.**

$$NDCG@K = DCG@K/IDCG@K, \quad DCG@K = \Sigma[r=1 \text{ to } K]\ (1[rel(d\_r)]) / \log_2(r+1) \tag{8}$$

5. Experimental Setup

The 500-QA pharmacovigilance benchmark was constructed in two stages. A PV-trained domain expert authored 150 seed questions spanning FAERS case interpretation, ICH regulatory timelines (E2A, E2B, E2C), causality methodology (WHO-UMC and Naranjo criteria), and MedDRA terminology; for each question the expert identified ground-truth passage indices directly from the corpus, ensuring labels reflect read source content rather than inferred embeddings. A further 350 pairs were generated by GPT-4o with prompts specifying clinical domain, query type, and expected answer scope, then reviewed by the same expert; 22 pairs with factual errors or ambiguous references were discarded, leaving 478 accepted pairs in total. Important caveats: (a) single-expert annotation without inter-annotator agreement reporting limits reliability assessment; (b) the 73% GPT-4o-generated component may introduce stylistic bias toward LLM-generated phrasing, potentially inflating ROUGE-1 and BLEU-4 scores for LLM-generated system outputs relative to what would be observed on fully human-authored references. Future benchmark versions should include multi-annotator validation and report ROUGE-1 separately for human-authored and GPT-4o-generated subsets. One hundred items were held out as the test set (seed 42), stratified across four query categories; the remaining 378 are reserved for fine-tuning.

All experiments ran on an NVIDIA Tesla T4 GPU (16 GB VRAM, CUDA 12.1) via Lightning.ai with 32 GB RAM. BGE-large-en-v1.5 (335M parameters) and ms-marco-MiniLM-L-6-v2 (22M parameters) ran in float16. Cross-encoder inference used batch_size = 1, which is a sub-optimal implementation for throughput; batched inference would substantially reduce reranking latency at the cost of implementation complexity. FAISS 1.7.4 managed index operations; IVFFlat k-means used seed 0. *rank-bm25* 0.2.2 provided BM25Okapi scoring with default parameters $k_1 = 1.5$, $b = 0.75$ (not tuned). Generation used boto3 1.35 against Amazon Bedrock Nova Lite (us-east-1, model ID: *amazon.nova-lite-v1:0*). Evaluation used *rouge-score* 0.1.2 and NLTK 3.8.1. Per-sample atomic checkpointing protected against data loss from mid-run session token expiry.

6. Results

6.1 Retrieval Quality

Table 1 reports retrieval metrics across all three thresholds. All values are point estimates on 100 queries; the 4-query Hit@5 gap between TurboRAG (1.000) and SimpleRAG (0.960) at $\tau = 0.65$ corresponds to exactly 4 discordant binary outcomes, which a McNemar test on 100 pairs yields as non-significant ($p \approx 0.13$) — an important caution against overinterpreting single-run results on small test sets. Nonetheless, TurboRAG leads every row at every threshold without exception across 15 data points, a consistent directional pattern. At $\tau = 0.65$, TurboRAG achieves perfect Hit@5 = 1.000 — every test query has at least one relevant passage in the top five. SimpleRAG misses four queries on this criterion. The practical implication scales with deployment volume: at high query rates, missed context leads to plausibly worded but unsupported responses, a qualitative failure distinct from a lower n-gram score.

Table 1. Retrieval metrics across thresholds (100 test samples, single run). ↑ = higher is better; bold = best value per row. S1 = SimpleRAG, S2 = OptimisedRAG, S3 = TurboRAG.

τ	Metric	S1	S2	S3
0.65	Hit@5 ↑	0.960	0.940	**1.000**
	Recall@5 ↑	0.830	0.802	**0.900**
	NDCG@5 ↑	0.825	0.805	**0.898**
	MRR ↑	0.862	0.869	**0.953**
	MAP ↑	0.789	0.845	**0.959**
0.70	Hit@5 ↑	0.800	0.790	**0.940**
	Recall@5 ↑	0.598	0.564	**0.708**
	NDCG@5 ↑	0.620	0.593	**0.710**
	MRR ↑	0.738	0.749	**0.834**
	MAP ↑	0.566	0.599	**0.776**
0.75	Hit@5 ↑	0.680	0.670	**0.860**
	Recall@5 ↑	0.402	0.350	**0.502**
	NDCG@5 ↑	0.447	0.403	**0.529**
	MRR ↑	0.657	0.654	**0.765**
	MAP ↑	0.383	0.381	**0.562**

Note. τ = embedding similarity relevance threshold; same BGE encoder is used for both retrieval and relevance labelling (circularity limitation). K = 5 reported as primary cutoff; results at K ∈ {1, 3, 10} follow the same directional pattern. Bold marks the highest value per row. TurboRAG leads all 5 metrics at all 3 thresholds (15/15 data points). NDCG@K = Normalised Discounted Cumulative Gain; MRR = Mean Reciprocal Rank; MAP = Mean Average Precision.

Threshold sensitivity. TurboRAG's NDCG@5 falls 41.1% relative from 0.898 (τ = 0.65) to 0.529 (τ = 0.75); SimpleRAG drops 45.8% (0.825 to 0.447). This is an evaluation artefact: retrieved passages are identical at all thresholds, but the stricter boundary excludes paraphrastic FAERS descriptions of the same clinical event whose BGE cosine similarities cluster between 0.68 and 0.73. Since τ classifies passages post-hoc without changing what is retrieved or generated, the 41% retrieval shift against <0.01 generation shift (ROUGE-1) exposes threshold selection as a measurement confound. System ranking is preserved exactly at all three thresholds, confirming that architectural conclusions from τ = 0.65 are directionally robust.

6.2 Generation and Context Quality

Table 2 presents generation and context metrics at τ = 0.65. TurboRAG leads on all 7 generation metrics (ROUGE-1/2/L, BLEU-4, METEOR, Token F1, Semantic Similarity). **However, as context budgets differ — SimpleRAG forwards 5 passages, the hybrid systems 10 — these generation gaps capture both retrieval quality and context-size effects jointly and should not be attributed to retrieval architecture alone.** The ROUGE-1 gap of 0.080 corresponds to approximately 8 additional matched unigrams per 100-word response, a meaningful difference for signal assessment memos where regulatory terminology alignment matters — but whether this is driven by retrieval or context size requires a controlled k=5-for-all experiment. Generation metrics are stable across all three thresholds: ΔROUGE-1 ≤ 0.007 for all systems, confirming that τ does not affect generator output.

Context metrics reveal a different pattern. SimpleRAG achieves the highest Context Precision (0.759), as dense-only retrieval concentrates the context window on geometrically query-close passages. OptimisedRAG achieves the highest Context Recall (0.811) because hybrid retrieval's broader evidence sweep covers more reference answer content. TurboRAG trails on both context metrics (Precision: 0.689; Recall: 0.782), reflecting the trade-off between TurboRAG's superior rank completeness (MAP = 0.959) and the dilution of query-centric precision when ten semantically diverse passages are concatenated.

Table 2. Generation and context quality at τ = 0.65 (100 samples, single run). Bold = best value per row.

Metric	S1 (SimpleRAG)	S2 (OptimisedRAG)	S3 (TurboRAG)
ROUGE-1 ↑	0.391	0.414	**0.471**
ROUGE-2 ↑	0.183	0.207	**0.262**
ROUGE-L ↑	0.282	0.313	**0.357**
BLEU-4 ↑	0.083	0.101	**0.146**
METEOR ↑	0.315	0.338	**0.395**
Token F1 ↑	0.345	0.366	**0.432**
Semantic Similarity ↑	0.862	0.868	**0.872**
Context Recall ↑	0.807	**0.811**	0.782
Context Precision ↑	**0.759**	0.735	0.689

Note. TurboRAG leads on 7 generation metrics (rows 1–7); SimpleRAG leads Context Precision; OptimisedRAG leads Context Recall. Generation metric stability: ΔROUGE-1 ≤ 0.007 and ΔSemantic Similarity ≤ 0.003 across τ ∈ {0.65, 0.70, 0.75} for all three systems — threshold choice does not affect generation output. Context metrics use the same BGE encoder as retrieval (circularity bias; see Section 4). Generation comparisons between SimpleRAG (k=5) and hybrid systems (k=10) are confounded by context budget; these differences should not be attributed to retrieval architecture alone.

6.3 Latency Decomposition

Table 3 reports mean latency by component. Generation latency is 1,099 ± 6 ms across all three systems, entirely from the Bedrock API round-trip. Generation represents 97.2% of SimpleRAG's total, 47.7% of OptimisedRAG's, and 31.5% of TurboRAG's total latency. All variability in total latency is retrieval-driven. SimpleRAG's 32 ms retrieval decomposes into ~28 ms BGE query encoding and ~4 ms flat-index scan. OptimisedRAG's 1,195 ms retrieval is dominated by cross-encoder inference (~1,140 ms for 100 candidate pairs at batch_size = 1, ~11.4 ms per pair). TurboRAG's 2,394 ms retrieval exceeds OptimisedRAG's despite identical downstream stages (both rerank approximately 100 RRF-merged candidates); the additional overhead likely arises from IVFFlat's non-contiguous memory access across 16 Voronoi partitions. Per-component profiling within the retrieval stage is needed for precise attribution. Latency values are single-run means; systematic variation across query types and document distributions is not characterised. Batching cross-encoder inference (e.g., batch_size = 32) would substantially reduce the 1,140 ms reranking cost and is a straightforward implementation improvement.

Table 3. Mean latency decomposition in milliseconds (τ = 0.65, 100 samples, single run). R:G = retrieval-to-generation time ratio.

System	Retrieval (ms)	Generation (ms)	Total (ms)	R:G	Gen. share (%)
S1 SimpleRAG	**32**	1,103	**1,135**	1:34	**97.2**
S2 OptimisedRAG	1,195	1,091	2,286	1.1:1	47.7
S3 TurboRAG	2,394	1,103	3,497	2.2:1	31.5

Note. Bold = best (lowest) value per column for Retrieval and Total; S1 also achieves the best Gen. share. Generation latency is constant at 1,099 ± 6 ms across all systems. Latency is hardware- and implementation-specific (T4 GPU, Lightning.ai cloud, cross-encoder batch_size=1); batched inference would substantially reduce cross-encoder overhead. All values are single-run means.

[Figure 2 — Insert image here]

Figure 2. System comparison at τ = 0.65. Panel (a): retrieval quality — Hit@5, NDCG@5, MRR, MAP, Recall@5. Panel (b): generation and context quality — ROUGE-1, ROUGE-2, METEOR, Token F1, Semantic Similarity, Context Recall, Context Precision. SimpleRAG (blue), OptimisedRAG (green), TurboRAG (orange). TurboRAG leads all 7 generation metrics and all 5 retrieval metrics. SimpleRAG achieves the highest Context Precision; OptimisedRAG achieves the highest Context Recall. Data from Tables 1–2.

7. Discussion

7.1 Deployment Profiles

Overnight FAERS batch processing tolerates any retrieval latency, making TurboRAG the natural choice: perfect Hit@5 eliminates context-absent failures, and the 0.080 absolute ROUGE-1 advantage over SimpleRAG (reflecting both retrieval quality and context-budget differences) directly improves signal assessment memo quality. Interactive reviewer tools with a 1.5-second hard budget require SimpleRAG — its 1,135 ms total falls within target while OptimisedRAG (2,286 ms) and TurboRAG (3,497 ms) do not. SimpleRAG's 4% Hit@5 shortfall in interactive settings can be mitigated via a UI fallback that triggers hybrid retrieval on low-confidence dense queries. On-premise deployment with a quantized local model (e.g., Llama-3.1-8B-Q4 at ~150 ms inference on the T4) would bring TurboRAG's total to approximately 2.5 s — viable for assisted-review workstreams with data residency constraints.

7.2 Why BM25 Matters in Pharmacovigilance

Removing BM25 from the hybrid pipeline costs 0.080 ROUGE-1 and four Hit@5 queries at τ = 0.65. The mechanism is structural: MedDRA Preferred Terms and ICH section identifiers are low-frequency tokens with high IDF weight in BM25. A query for 'rhabdomyolysis' or 'ICH E2A Section 3.2 criterion A' can surface ground-truth passages through BM25 that dense search misses, because the MedDRA PT competes in embedding space with clinical synonyms ('muscle breakdown', 'myoglobinuria'). The 0.4/0.6 BM25-to-dense weighting formalises this complementarity, though systematic weight optimisation on a validation set is recommended.

7.3 Ethical and Safety Considerations

Three concerns govern responsible deployment. First, retrieval grounding substantially but not completely eliminates hallucination; any generated output contributing to regulatory submissions must undergo expert pharmacovigilance review. Ball et al. established that trustworthiness and explainability — not performance metrics — determine AI tool acceptance by FDA safety reviewers [29]; source-attributed generation is therefore a minimum requirement. Second, FAERS reflects systematic reporting biases by drug class, patient demographics, and reaction severity; retrieval systems trained on FAERS inherit these biases. Third, even the public FAERS release carries re-identification risk; any extension to proprietary case data requires formal data governance review.

7.4 Limitations

Seven limitations bound these conclusions. (1) *Context budget confound:* SimpleRAG forwards 5 passages to the generator; the hybrid systems forward 10. All generation metric differences between S1 and S2/S3 are jointly determined by retrieval quality and context size; an experiment equalising k across systems is needed to isolate retrieval effects. (2) *Embedding circularity:* Context Recall, Context Precision, and Semantic Similarity use the same BGE encoder as retrieval; results are potentially biased toward dense-retrieval behaviour; independent evaluation would mitigate this. (3) *Small test set with no statistical testing:* 100 queries without CIs or significance tests means reported improvements may not be robust; the 4-query Hit@5 gap is non-significant by McNemar's test. (4) *Corpus scale:* 19,930 passages is small relative to the full FAERS archive; IVFFlat scalability requires separate testing. (5) *Dataset construction:* 73% GPT-4o augmentation without inter-annotator agreement; GPT-4o phrasing may inflate lexical metrics for LLM-generated system outputs. (6) *No faithfulness evaluation:* n-gram and embedding metrics are weak proxies for factual correctness in safety-critical contexts [29]; citation-grounded evaluation (QAFactEval, attribution metrics) is a high-priority extension. (7) *Latency characterisation:* latency values are single-run means at batch_size=1 cross-encoder inference; they are hardware-specific, not architectural constants, and batching would substantially reduce the cross-encoder bottleneck.

8. Conclusion

To the best of our knowledge, this study provides the first controlled comparison of dense-only, hybrid, and IVFFlat-index hybrid RAG architectures on a pharmacovigilance corpus. Three research questions are answered. (RQ1) TurboRAG leads on all 5 retrieval quality metrics and all 7 generation quality metrics (12 of 17 reported metrics) at $\tau = 0.65$, confirming that BM25-dense fusion materially improves over dense-only retrieval for PV question answering; the additional advantage of TurboRAG over OptimisedRAG is hypothesised to reflect IVFFlat's locally coherent candidate pools but requires ablation to confirm. SimpleRAG leads Context Precision and both latency metrics; OptimisedRAG leads Context Recall — no system is universally superior. (RQ2) Tightening τ from 0.65 to 0.75 collapses retrieval metrics by up to 41% relative while ROUGE-1 changes by less than 0.01, exposing threshold selection as a measurement confound that PV-RAG evaluations should standardise. (RQ3) API generation (~1,090 ms) dominates total response time for all three systems, making context rather than retrieval speed the primary deployment decision variable. Priority future work: equalise context budgets (k=5 vs k=10) to isolate retrieval effects; add bootstrap CIs and significance testing; introduce citation-grounded faithfulness evaluation; extend to full FAERS scale; and benchmark on-device generation to eliminate the API latency bottleneck.

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11. Data Availability Statement

The 500-QA pharmacovigilance benchmark will be released on GitHub upon publication acceptance. FAERS source data are publicly available at https://www.fda.gov/drugs/fda-adverse-event-reporting-system-faers. Experimental code will be released simultaneously.

12. Conflict of Interest

The authors declare no conflict of interest.

13. References

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